THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 8: **Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A blood sample has 5 million RBC per mm³. In 2 mm³ the number is: 1

(a) 5 million **(b)** 2.5 million **(c)** 7 million **(d)** 10 million

2. Which of the following explains the red colour of blood? 1

(a) insulin **(b)** bile
(c) the iron-containing haemoglobin **(d)** chlorophyll

3. If the diaphragm fails to contract: 1

(a) digestion stops **(b)** heart stops
(c) inhalation is affected **(d)** kidneys stop

4. A person passes about 1.5 L of urine per day. In a week, the amount is: 1

(a) 1.5 L **(b)** 105 L **(c)** 15 L **(d)** 10.5 L

5. Which statement correctly relates structure and function? 1

(a) capillaries have valves
(b) veins have thick walls
(c) arteries have thick walls to bear high pressure
(d) arteries have valves

6. A person climbing a high mountain breathes faster because: 1

(a) air is hot **(b)** he is hungry
(c) air is wet **(d)** thin air has less oxygen

7. A cow chews cud for 8 hours a day. In a 30-day month this is: 1

(a) 8 hours **(b)** 24 hours **(c)** 38 hours **(d)** 240 hours

8. In the lungs, blood gains oxygen and loses: 1

(a) bile **(b)** water entirely **(c)** nitrogen **(d)** carbon dioxide

9. **Assertion (A):** Arteries carry blood towards the heart. 1
Reason (R): Veins carry blood towards the heart.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Haemoglobin in red blood cells carries oxygen. 1
Reason (R): Plasma is the liquid part of blood.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are the lungs spongy? 2

12. Why is fibre in the diet important for the large intestine? 2

13. Explain how the lack of an enzyme (like lipase) would affect digestion. 2

14. Why does a person with kidney stones need to drink more water? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why we feel hungry again after a short time on eating only sweets. 3

16. Explain the process of peristalsis and its importance. 3

17. Explain how human beings maintain a constant body temperature using respiration and sweating. 3

OR

A person's urine is very dark and small in volume. What could be the reason and advice? 3

18. Why do athletes have a larger lung capacity? 3

OR

Why is mother's milk the best food for a baby? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sneha runs a 100 m race at her school in Kolkata. At the end, her heart beats faster and she breathes heavily. Her teacher says that her blood is carrying more oxygen to the muscles. 5
(i) Name the organ that pumps blood.
(ii) Name the pigment in red blood cells that carries oxygen.
(iii) Where does exchange of gases take place in the lungs?
(iv) Why do breathing and heartbeat rates increase after running?

20. Discuss the relationship between the digestive, circulatory and respiratory systems in releasing energy. 5

OR

Describe how first-aid for a bleeding wound uses ideas about blood clotting and infection. 5

END OF QUESTION PAPER — **Total: 40 Marks**